



Newsworthify me, positively or negatively:
a multimodal approach to examine media frames during 2023 Dutch parliamentary elections



This was my very first conference presentation, held on Thursday, February 2 in Rotterdam, the Netherlands. The project is part of the NWO VICI project on discontent framing, supervised by prof.dr. Vliegenthart, prof.dr. Kruikemeier, and dr. Vermeer.

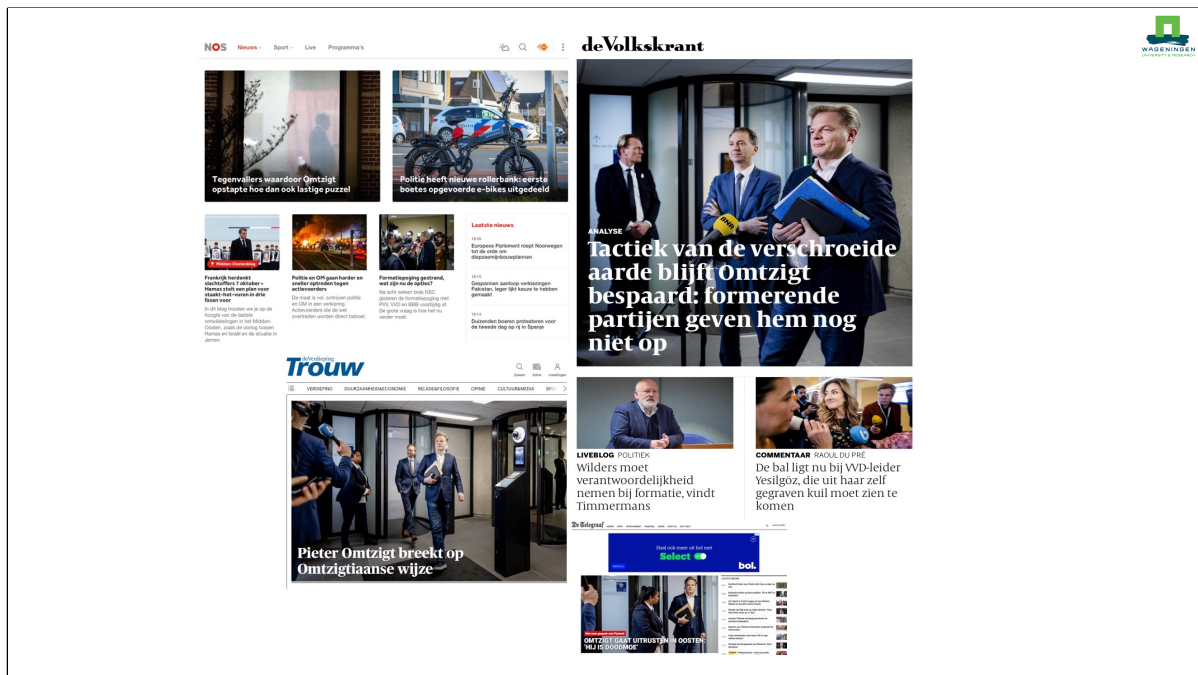
I take a moment to stress that at this point in time, the project was in its exploratory phase, with the aim to eventually study multi-modal frames by news media and politicians

This presentation is focused more on coverage than it is on framing, although framing is what we are eventually most interested in.

Let's start. I want to take a minute for a little experiment. I will soon say a word that might trigger some thoughts. I want you to remember the first thing that comes to your mind. Are you ready? Now close your eyes...

Politician

You can open your eyes now. For whom did this trigger a visual image? Whom thought of a politician currently in office? And what were the characteristics that you can attribute to the person that came to your mind? Was this person male or female? Intelligent? Witty? Charismatic? Furious? Serious?



Online media, be it news websites or social media platforms, are increasingly **multi-modal**.

And because our **biological attention** is naturally more noticable of visual over **textual information**, the **attention gathers** on these information channels are often visual.

As we've experienced in our little experiment, visual information triggers our mind to make associations – associations many of us are likely to recall from memory more easily than text.

Visual frame:
Distant,
unapproachable

Textual frame:
*Setbacks for which Omtzigt
resigned remain a challenging
puzzle nonetheless*





Visual frame:
Content,
confident,
ready to announce

Textual frame:
*Scorched earth policy remains
avoided for Omtzigt: parties in
negotiation not yet give up on
him*

Visual frame:
Tired,
overwhelmed by
journalists

Textual frame:
*Omtzigt breaks with
Omtzigtian manner*





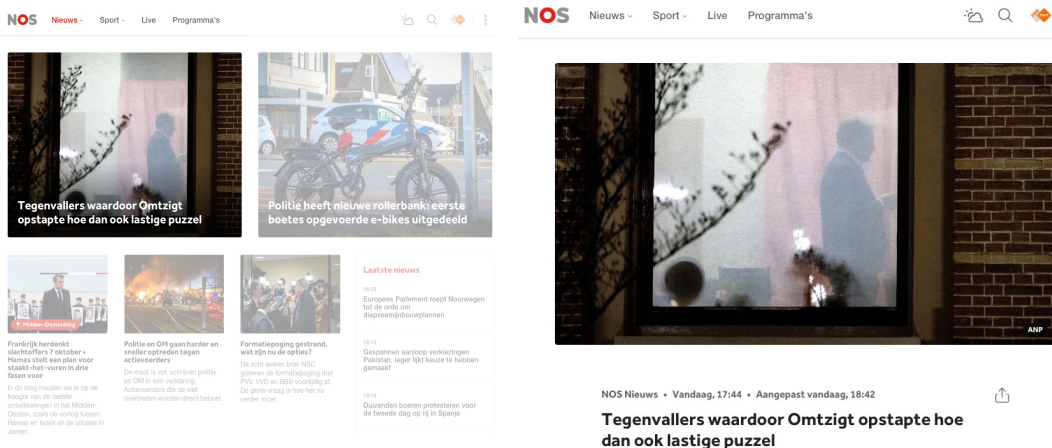
Visual frame:
Tired,
closed

Textual frame:
*Omtzigt is going to rest in the
East: 'he is dead tired'*

However, **when images and text get combined**, i.e. **multi-modal**, they are more powerful, as we can see on this slide.

Moreover, accompanying a visual frame of a politician with a general news image, may create a salience with this politician that is irrelevant, but impactful. However, some politicians, just like some news items about political issues, are simply more newsworthy than others.

NOS.nl's multi-modal frames during election period

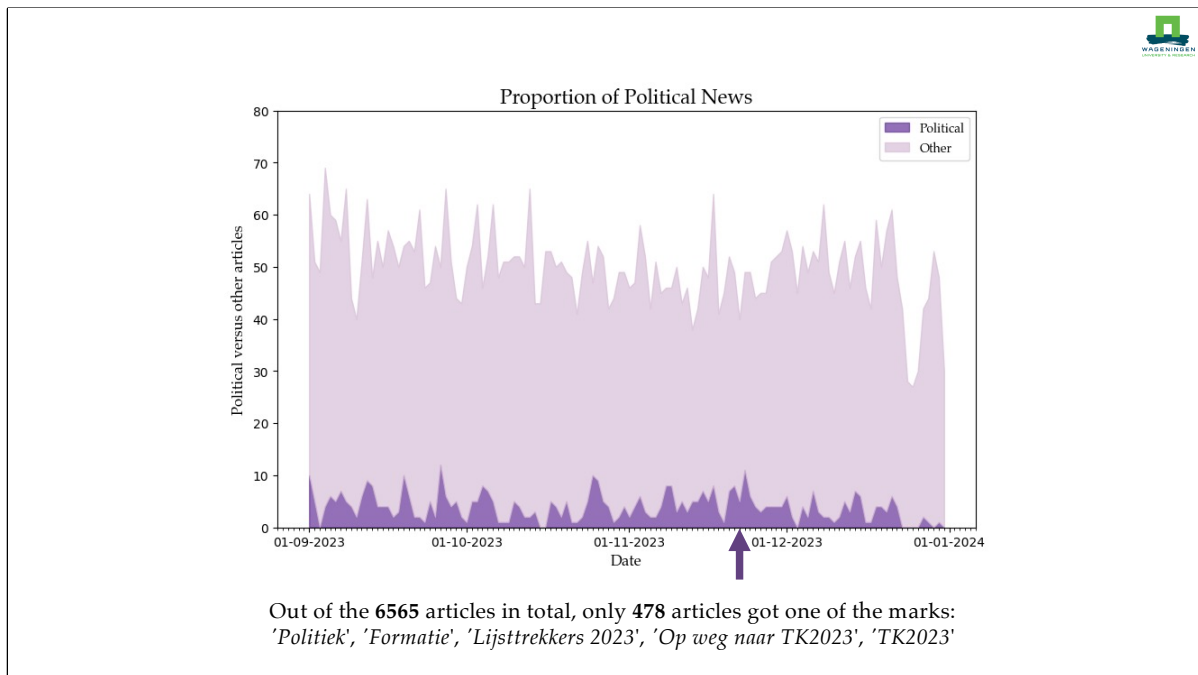


homepage

article page

In this study, we analyzed the visual frames of the main images that appeared with the Dutch public news outlet, NOS, news articles (i.e. the image tiles that are presented on the homepage of NOS.nl and are largely depicted when opening the article - all images that appeared later in the article were out of scope).

We want to find out whether some politicians get significantly more coverage, which may or may not be explained by their newsworthiness. We consider the politician newsworthy if the article of interest is about this politician, which we will judge by the textual title frame. We therefore seek correspondance of the visual and the textual frame. In the example on the slide, this correspondance holds.



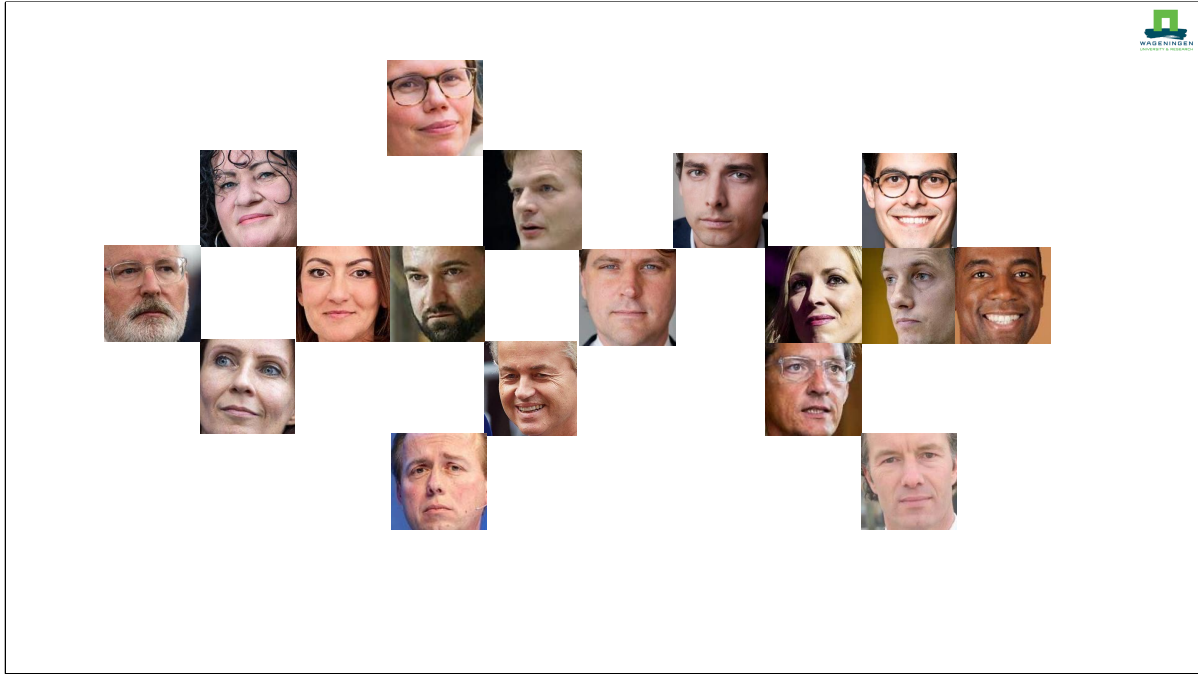
So let's first take a look at what we could gather with web scraping in Python. We set the time scope of the pre-, during and post-election period on September 2023 to December 2023. We found 6565 articles, with no strong trend visible at first glance.

Out of these articles, 478 got the 'category' label of ['Politiek', 'Formatie', 'Lijsttrekkers 2023', 'Op weg naar TK2023', 'TK2023']



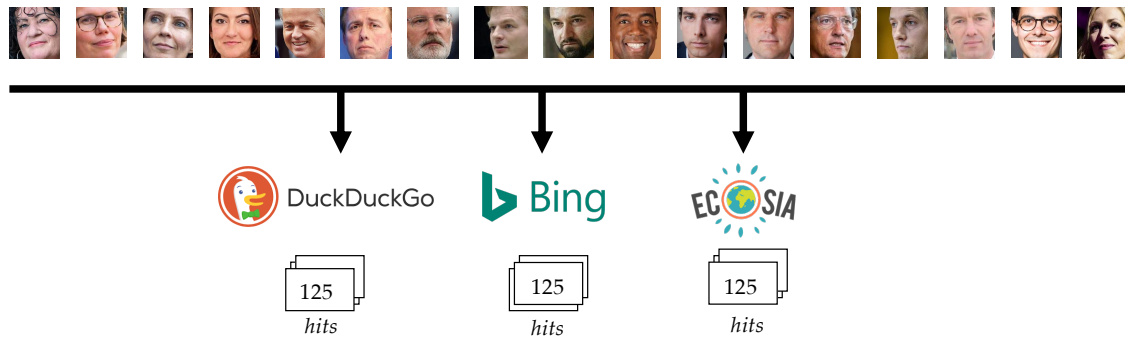
Let's go back to our case. We want to find the news photos on which politicians appear in a visual frame.

Now obviously, manually annotating 6565 news images is a pain in the ass. That's why we explored some cool methods introduced by colleagues in the field of computer sciences. It's called face recognition.

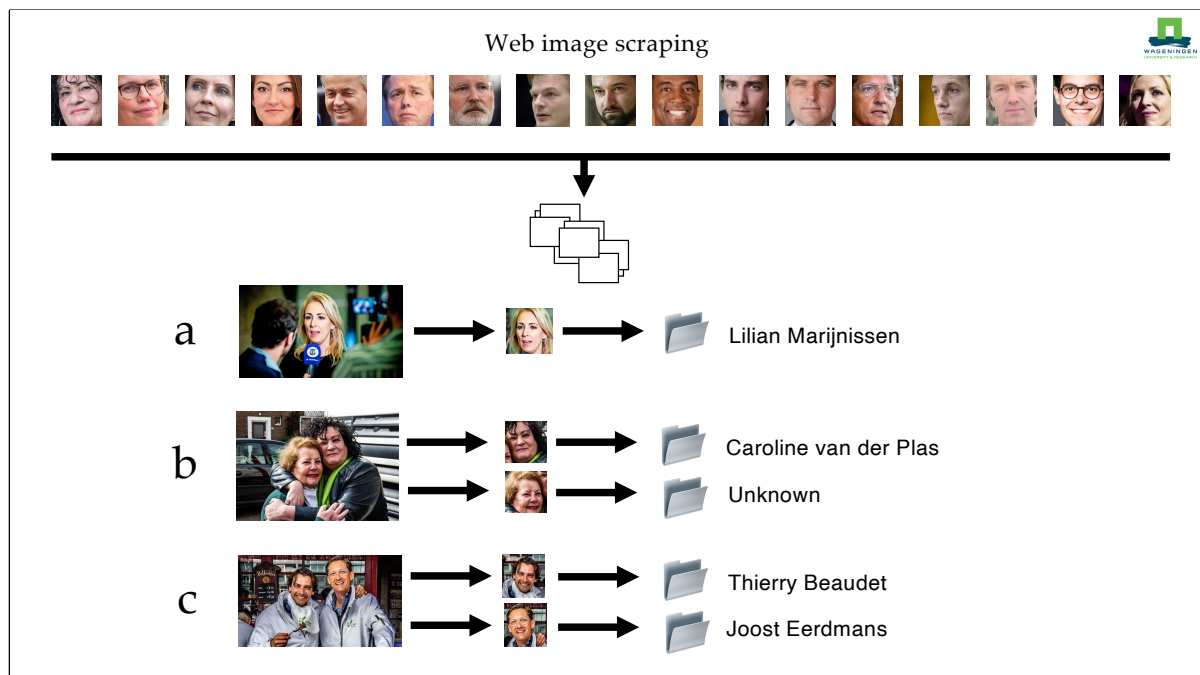


For the 17 political party frontmen and –women running for office, we gathered 125 images on 3 different search engines

Web image scraping



I refer to these with 'hits'. In order to train a model such that it is able to recognize a politician, you need to have input data where the photo of this politician and this politician alone gets the correct label. Now let me tell you, that the most random people end up between our hits. But not only random people: also other politicians. That's where it becomes tricky.

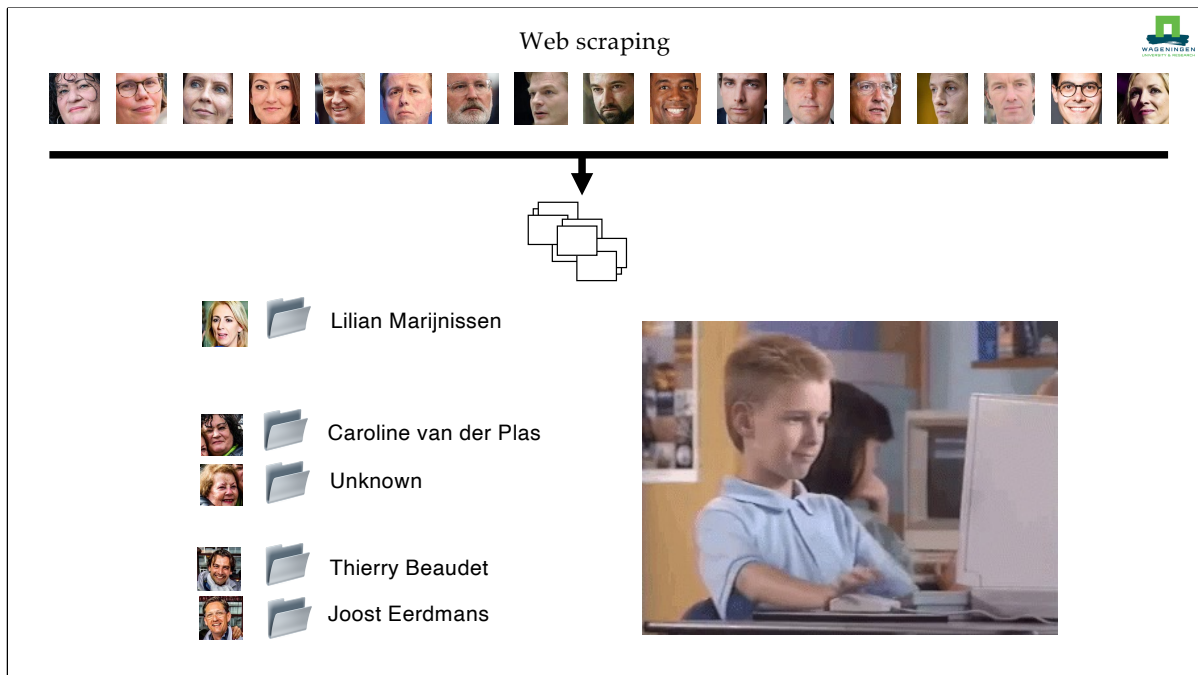


We isolated and cut out the faces using the Python library `face_recognition`, and manually assured a person was placed in the right folder.

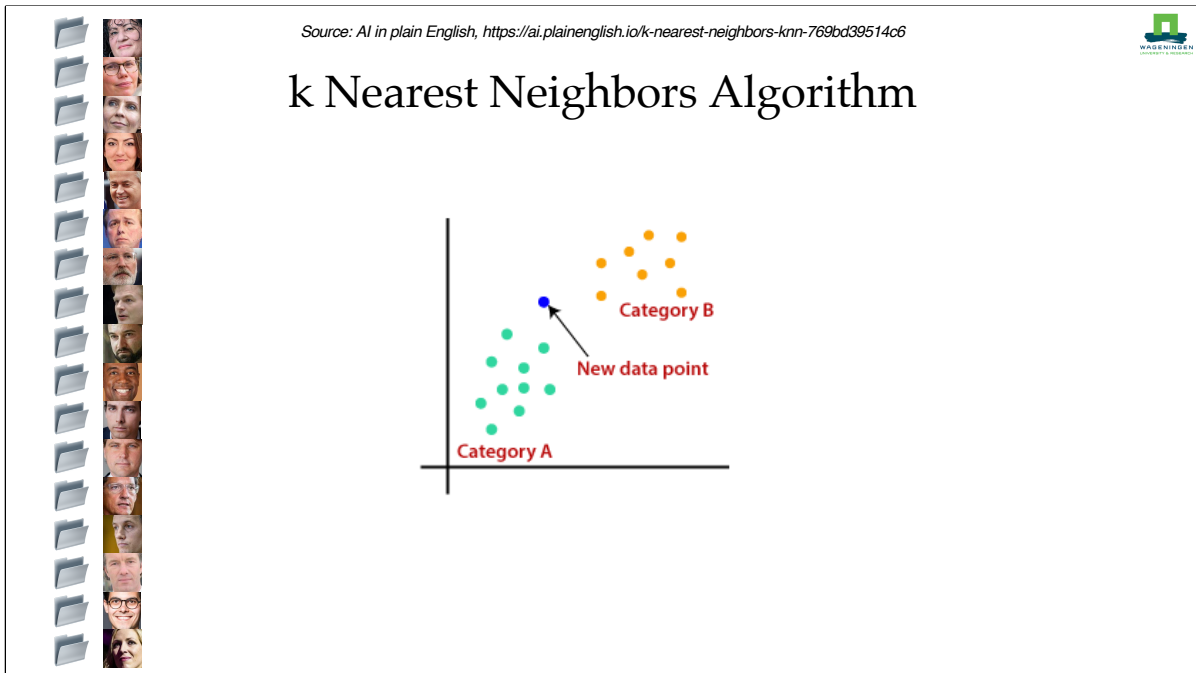
Case A shows the ideal case, where the politician of interest is the only one depicted and identifiable on our hit.

Case B shows an error case, where both the politician as well as a random person ended up on the hit. The random person was therefore doomed to the Unknown folder.

Case C shows a different error case, where two politicians of interest appear in each others' hit. If this error occurred, we brought the lost politician back home ;)



So far so good



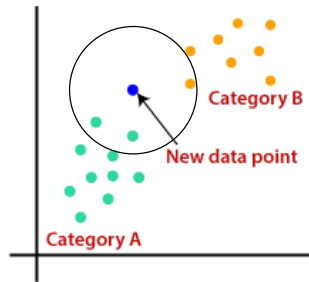
A very simple model to classify the images of politicians, is the k-Nearest Neighbors model. Because I personally have a strong preference for simple and explainable models.

I chose to explore the applicability of this model to the problem at hand. It's so simple that my cat understands it, and I don't even have a cat.

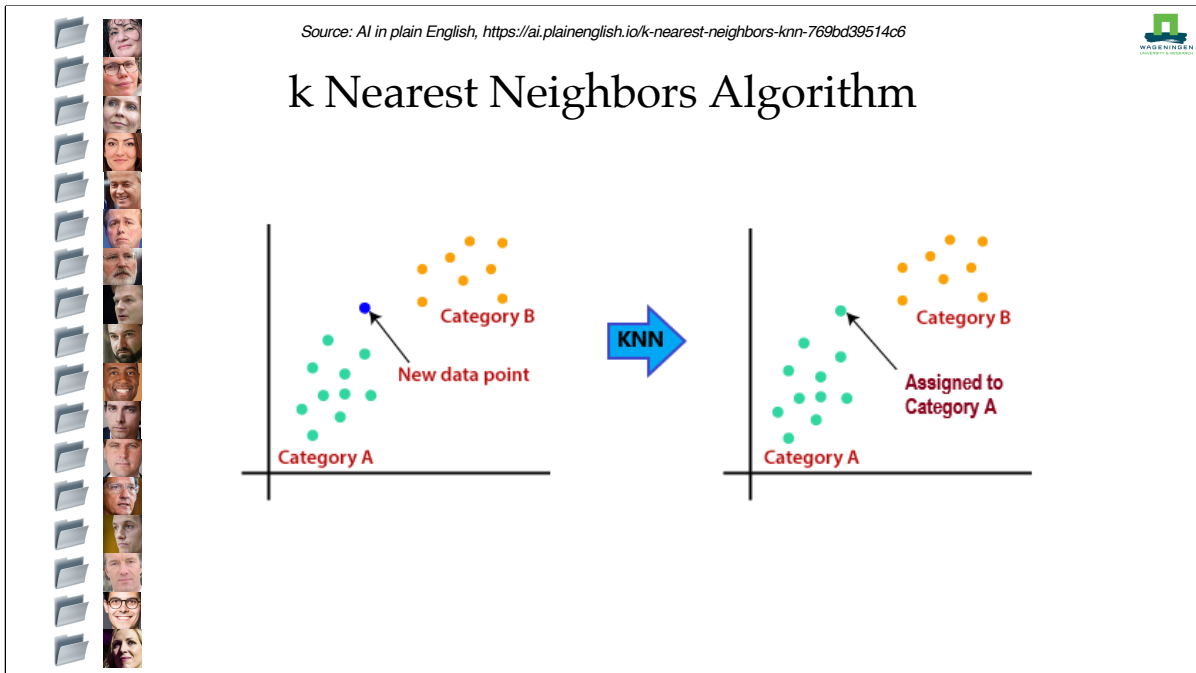
Source: AI in plain English, <https://ai.plainenglish.io/k-nearest-neighbors-knn-769bd39514c6>



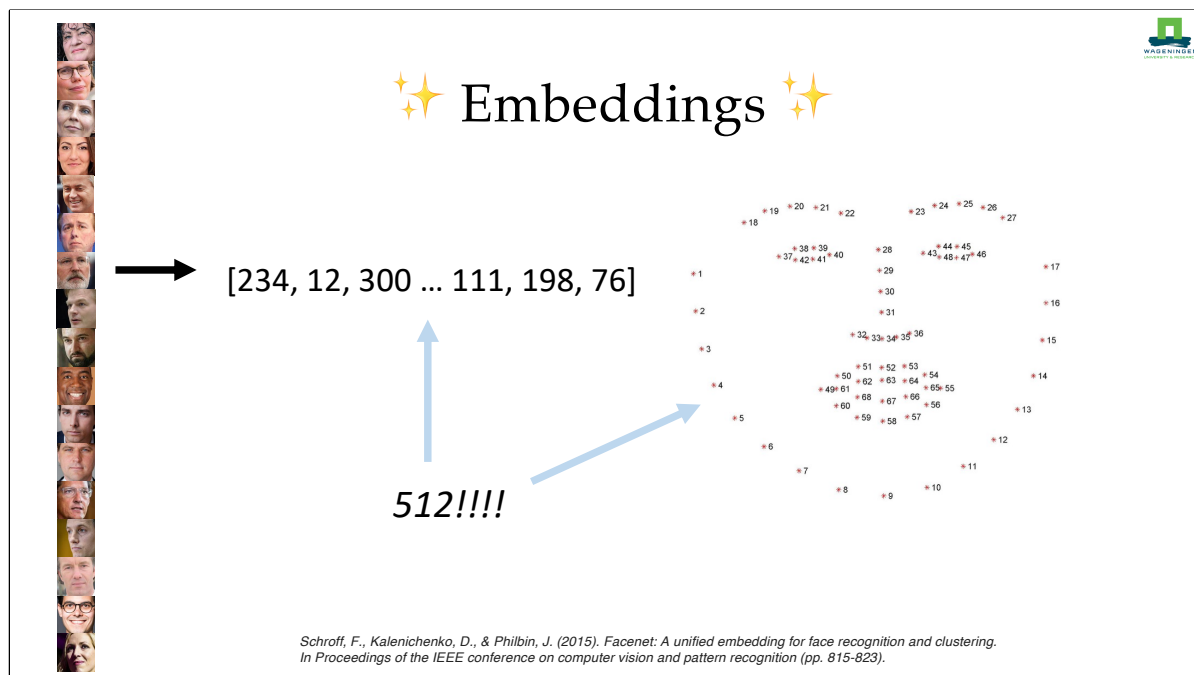
k Nearest Neighbors Algorithm



Every data point, or politician in our case, gets assigned a numerical representation in space, of course with the right label. Now every new, unseen politician that enters this space, will get the label assigned that is closest k neighbors. Now suppose we set the number of neighbors, k , to 2. What label would the new datapoint be assigned?



Exactly, category A!



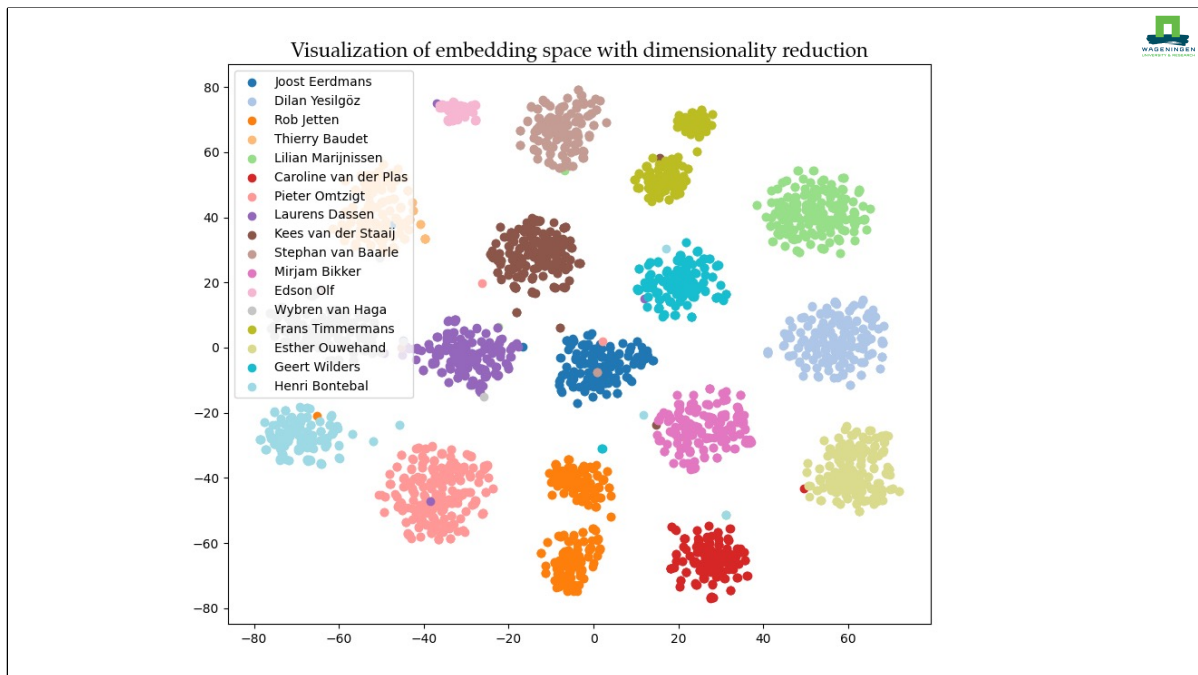
So how do we get from an image to a numerical representation? Embeddings.

For every politician, we use FaceNet (a mega sophisticated state-of-the-art model trained on tons of face images around the world) to calculate embeddings of their face images. That is, an image goes in, and a list of numbers come out.

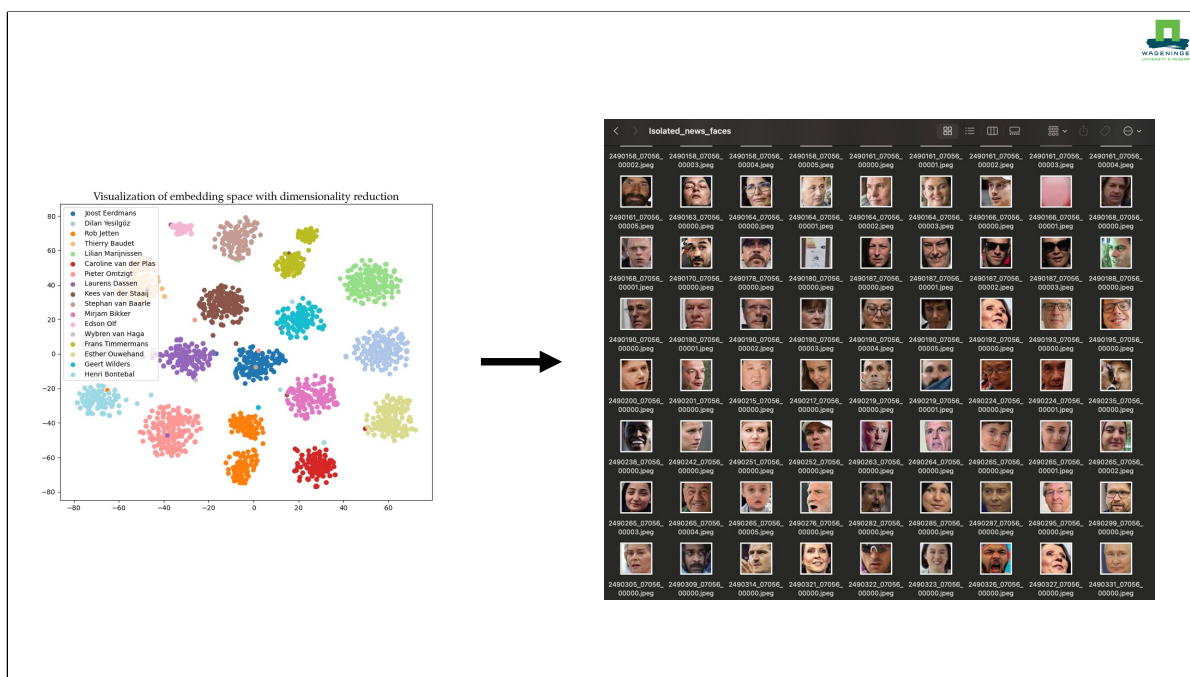
You can see embeddings as numerical recognition points, where eyebrows, noses, ears, eyes, and all that jazz get a value unique to the input data.

The number of embeddings is enormous: 512 per image. Whereas we are used to a 2-dimension coordinate for spatial location, our embeddings shape a 512-dimension coordinate.

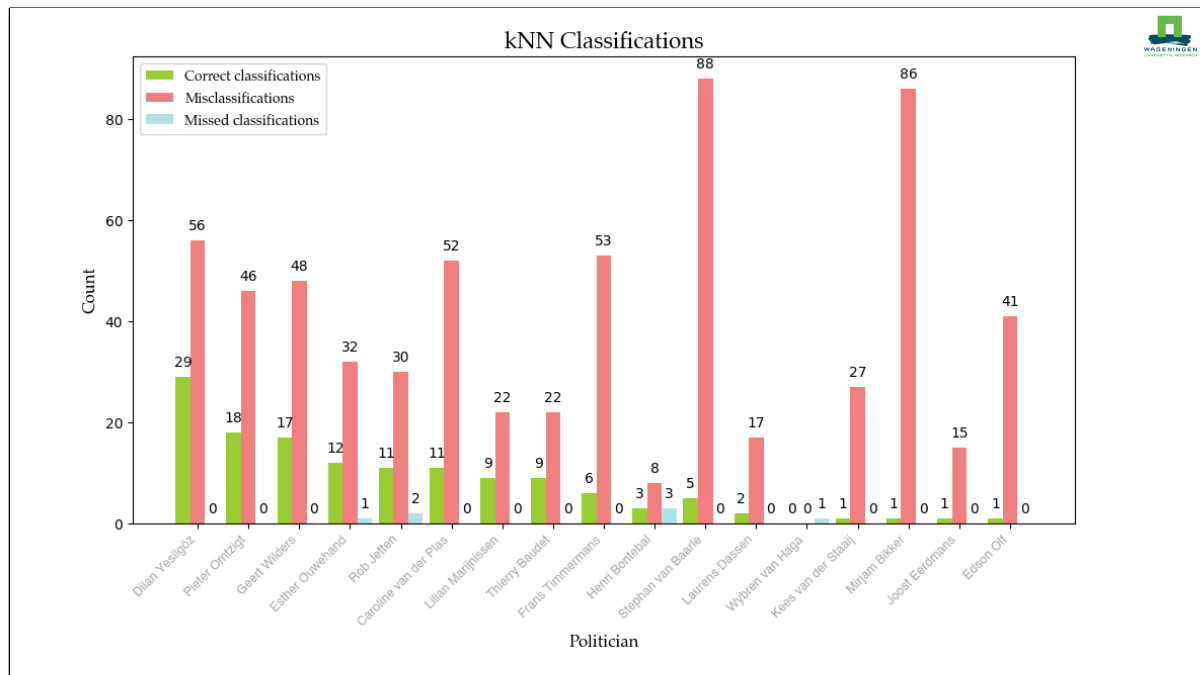
It's computationally impossible to show how that embedding space would look like, but luckily there exist dimensionality reduction techniques that are able to save as much information as possible while bringing down the number of dimensions. This yields the following result.



As you can see, even with a downscaled complexity, the politicians' faces cluster pretty well.



Let's find out if this works when applying it to the cut-out faces we found in news images that were published with NOS articles during the election period.



It doesn't ☺ We see many misclassifications (i.e., unknown people mistaken for politicians) and missed classifications, i.e., politicians doomed to the unknown folder.

This study is of a relative small scale, so to make sure the dataset is of best quality, we did another manual correction so make sure that the probability of missing a visual politician frame was minimalized.

Funky misclassifications



Caroline 'Korea' van der Plas



Kees 'Mighty Mike' van der Staaij



Geert 'Oranjeleeuwilders'

This manual correction was obviously a lot of fun as the most funky misclassifications came about

Funky misclassifications



Wait, hang on...



Funky misclassifications

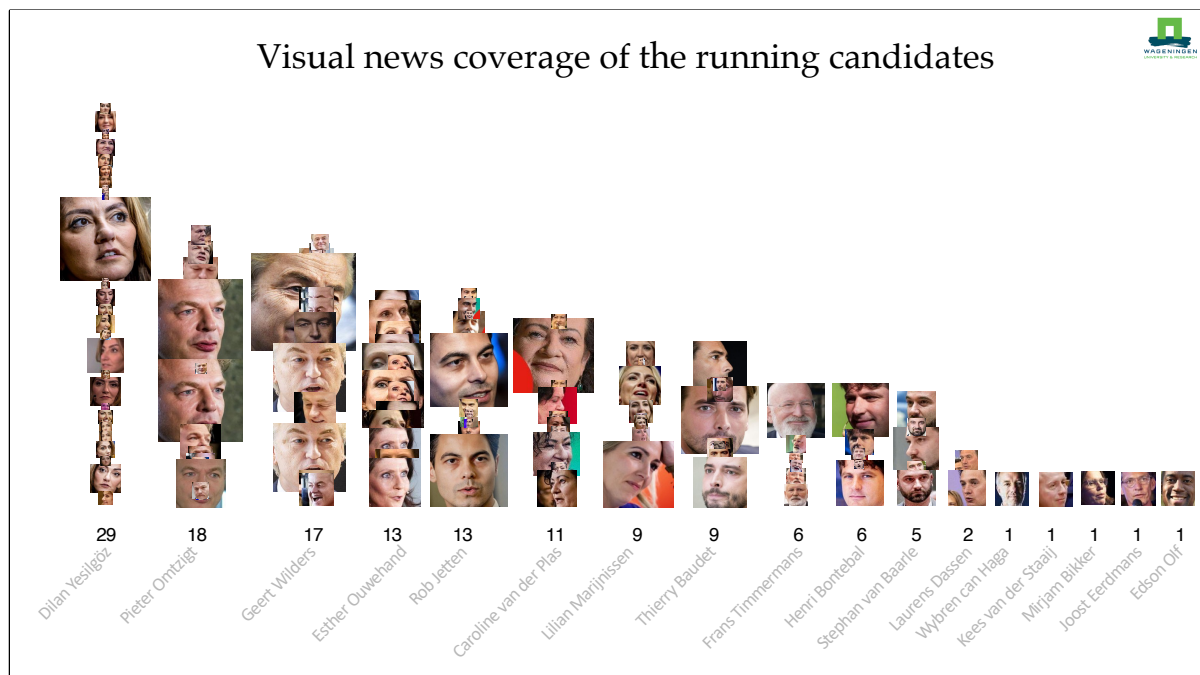


Wait, hang on...



Frans 'Kremlin OG'
Timmermans

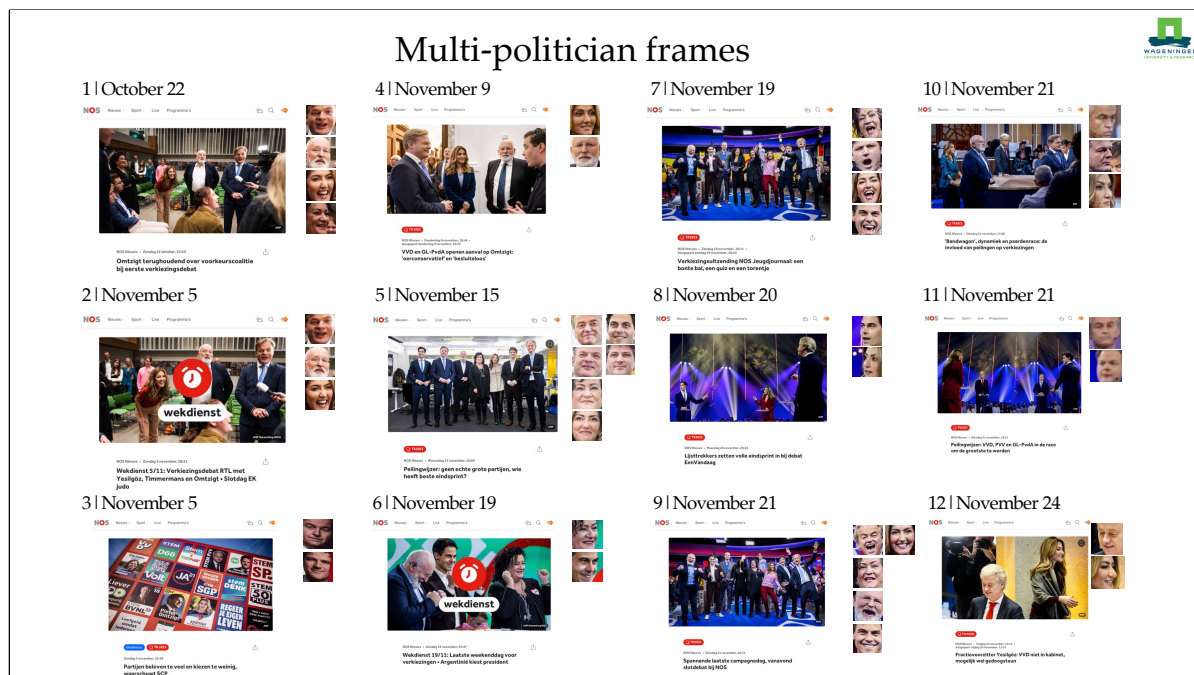
Frans, is that you?



We are now left with all the politician face images that we were able to find in the data. Let's take a closer look.

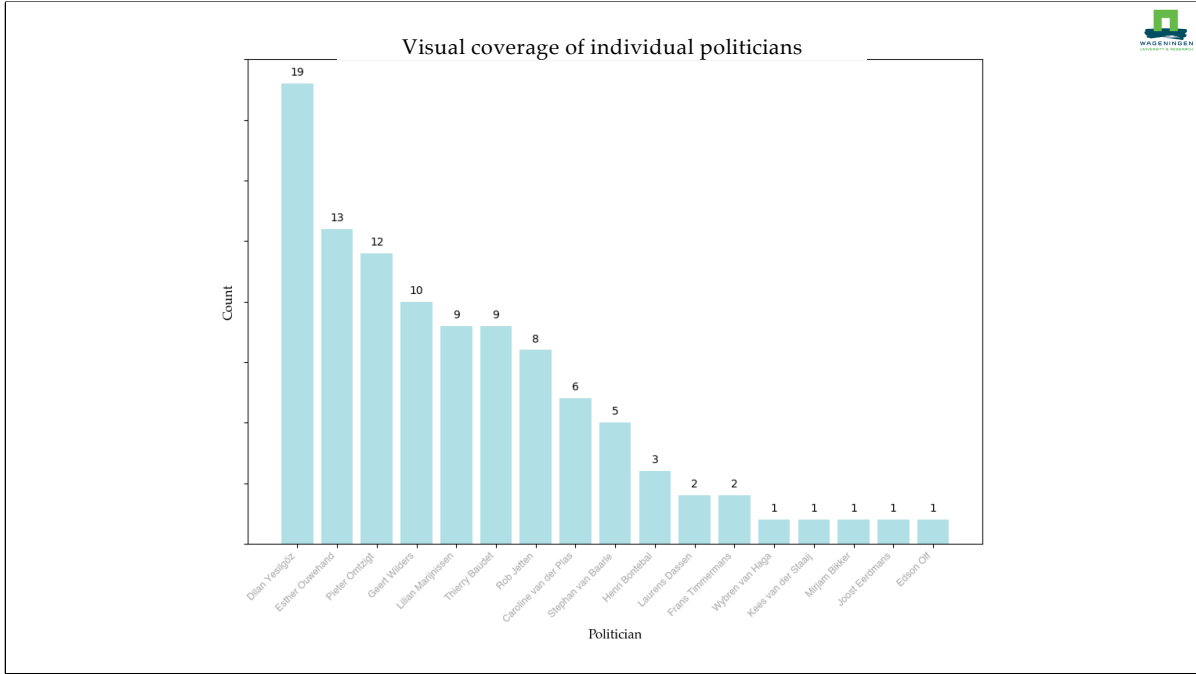
Some political party leaders running for office in the national elections enjoyed more visual presence than others, as we can see here. Especially Dilan Yesilgöz stands out.

We're not there yet, though. Some politicians appear on a visual frame together.



We see that in most cases all or a majority of the faces were identified by the python library, but definitely not always.

We also see some of NOS.nl's creativity, as they re-used photo 1 to photo 2, but cut out Caroline van der Plas to make it compliant with the textual frame. On photo 8, we see Geert Wilders' face was not spotted and was therefore not in the dataset hence not shown to the model. That is probably due to the angle: he is turned to the camera with his back.



If we disregard the shared politician visual coverage, we have the following individual coverage left.

When I explored this data combined with the textual title frames, although briefly, I noticed that for most of the articles that featured a visual politician frame, the textual title frame did indicate correspondance.

It is too early to draw conclusions, but it appears to me that NOS is doing a pretty good job at limiting their bias given the newsworthiness of some politicians over others.

I'd like to open the discussion on where we can go from here. To make the first contribution myself... because I haven't said enough yet, obviously... I'd like to go back to the start of the presentation, and more particularly to your visual associations with the word politician. What characteristics did you attribute mindfully to your associations? To what extent are they influenced by the media you consume? To what extent do you think they are biased?



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VICI

The mobilization of political dissatisfaction: causes, content, consequences

when. do they pick a single politician image frame

NOSA SOMETIMES changes headlines - what gets most clicks

WHEN does it influence my reaction?

image and later text --> evaluation process and image shaping in your mind

study interaction between image and frame